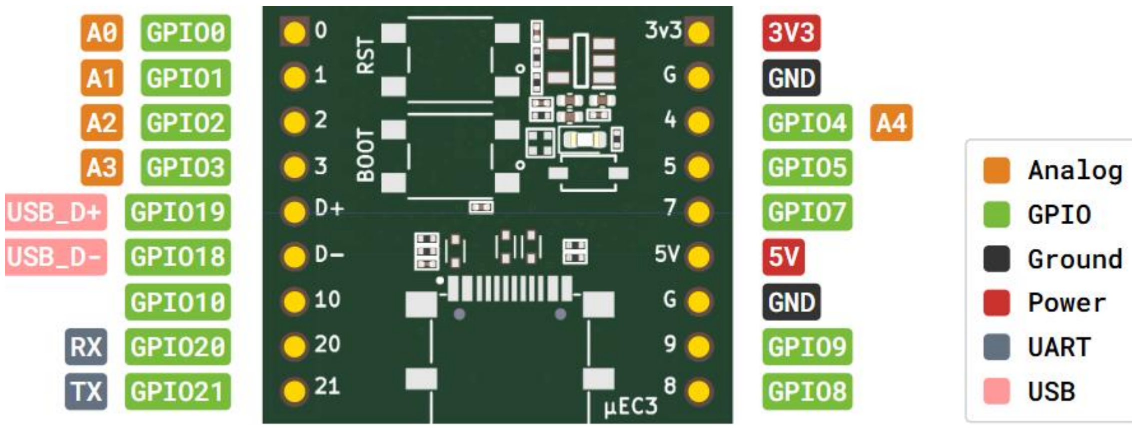
Micro-EC3: "ESP32-C3 WROOM 02" WiFi/Bluetooth Microcontroller Dev Board



****

Overview

Module: *ESP32-C3 WROOM 02 module*

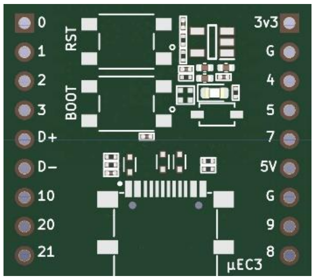
Applications: Low power IOT sensing, Wearables

Datasheet: documentation.espressif.com/esp32-c3-wroom-02_datasheet_en.pdf

Note: Any GPIO can be configured as I2C, I2S, PWM, UART and TWAI pins. GPIO 6 is configured as an onboard blink LED. GPIO 9 is configured as BOOT Pin with onboard 10K pull-up. GPIO18, 19 are dedicated USB data pins. GPIO20, 21 are UART RX and TX pins.

Specifications:

Board dimension	2.7 x 2.4 cm (Tiny coin sized)
Processor	32-bit single core RISC V microprocessor with max frequency 160 MHz
ROM/SRAM	384 KB ROM/400 KB SRAM on-chip
Flash Memory	4 MB
Wireless Communication	2.4 GHz Wi-Fi (802.11b/g/n), Bluetooth 5.0 (Bluetooth Low Energy)
IO Pins	14 GPIO pins, 5V power pin and 3.3V LDO output pin exposed.
Interface support	I2C, I2S, SPI, 12-bit ADC – 2 channels/5 Inputs, PWM, UART
Built-in LEDs	GPIO6 for control/blink LED. Separate LED is included for power indication
Antenna	On-board PCB antenna
USB Programming	USB-C port with native USB programming
Switches	On-board BOOT and RESET buttons



µEC3 (MicroEC3) - A Tiny ESP32-C3 Microcontroller Development Board

µEC3 Product Details

1. Overview

µEC3 is a high-performance, tiny microcontroller development board based on ESP32-C3 WROOM 02 module. The board runs on 32-bit RISC V single core processor with a maximum frequency of 160 MHz. The module has 4 MB flash memory and the board is specifically targeted for low power IOT sensing applications.

The module is equipped with PCB antenna with support for 2.4 GHz WiFi, Bluetooth LE communication. Board programming is done via on-chip/native USB without using an external USB-UART bridge chip.

With its high-performance module, the µEC3 board is tiny (2.7 x 2.4 cm) unlike other dev boards thereby making it highly suitable for space constrained IOT applications and wearables. µEC3 board supports firmware development on ESP-IDF, Arduino IDE, Micro Python/Circuit Python and other similar platforms.

2. Applications

µEC3 is an embedded IOT high performance device that is highly suited for low power applications such as:

- Smart homes
- Industrial automation
- Health care
- Consumer electronics
- Smart Agriculture
- Low-power IoT Sensor Hubs
- Low-power IoT Data Loggers

3. Board Specifications

Board dimension	2.7 x 2.4 cm
Processor	32-bit single core RISC V microprocessor
Max processor frequency	160 MHz
ROM/SRAM	384 KB ROM/400 KB SRAM on-chip
Flash Memory	4 MB
Wireless Communication	2.4 GHz Wi-Fi (802.11b/g/n), Bluetooth 5.0 (Bluetooth Low Energy)
IO Pins	14 pins with flexible IO configuration, 5V power pin and 3.3V LDO output pin
Interface support	I2C, I2S, SPI, 12-bit ADC - 2 channels/5 Inputs, PWM, UART
Built-in LEDs	GPIO6 for control/blink LED. A separate LED is included for power indication
Antenna	On-board PCB antenna
USB Programming	USB-C port with native USB programming
Switches	On-board BOOT and RESET buttons

---

4. μEC3 Product Image

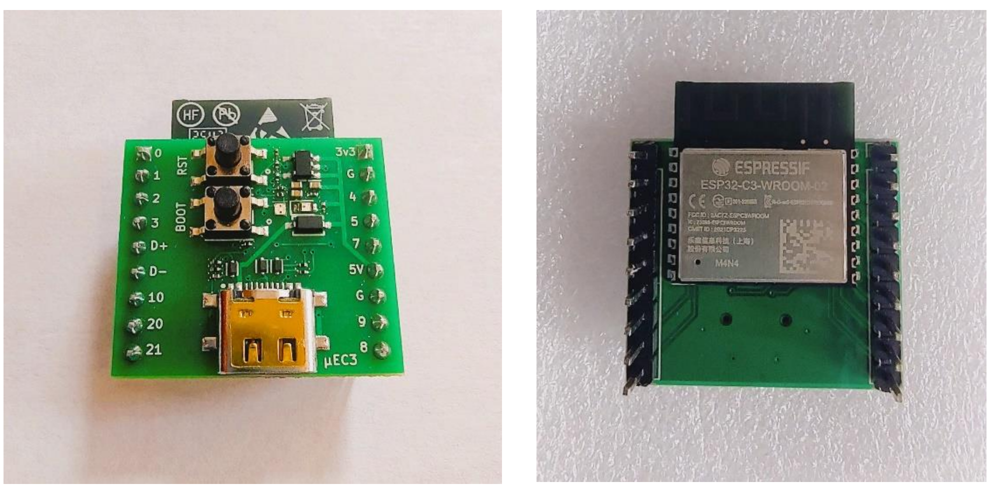


Fig. 1 Front and Back Images of μEC3 Board

5. Pinout Diagram

Board dimension: 2.7 x 2.4 cm

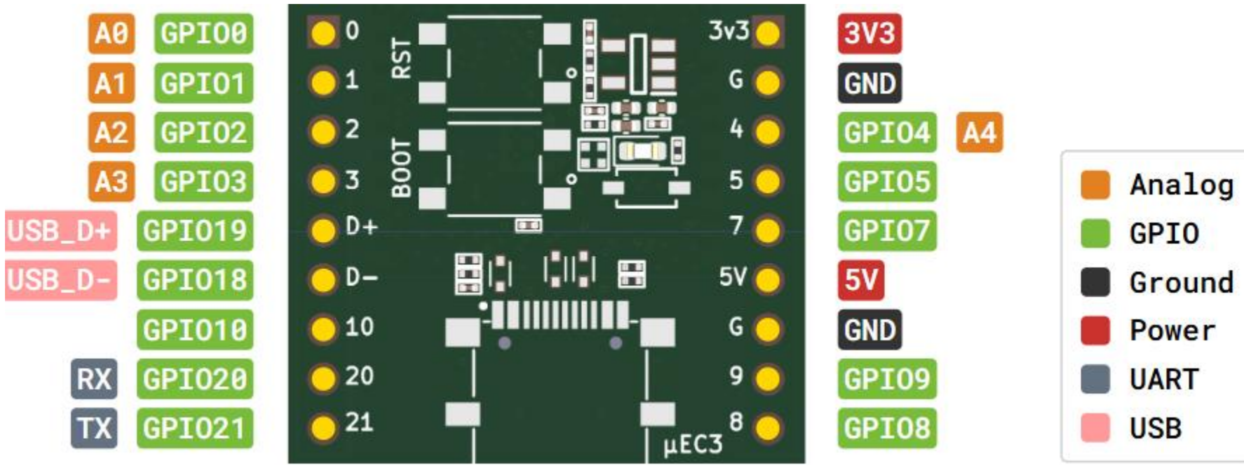


Fig. 2 µEC3 Pin-out illustration

6. Description

µEC3 has an on-board control LED which is connected to GPIO 6 for blinking and other control purposes. In addition to the control LED, there is a separate LED for power indication.

The current capability of the on-board LDO is 800 mA. The regulated output of the LDO is exposed as 3.3V pin on-board which can be used to drive the external circuits if needed. The on-board 5V pin is the USB power pin that is shielded from VBUS of USB-C port through a Schottky diode.

The USB-C port provides power as well as programming of the chip. There is no on-board USB-UART chip and hence the programming/flashing is done directly via the native USB peripheral of ESP32-C3 SOC.

Pinout details and functionality of all exposed pins are shown in Fig. 2.

GPIO 18 and GPIO 19 are USB data pins. GPIO 9 is the boot pin with an external pull-up of 10 kOhm resistor. If the above pins need to reconfigured as other GPIO pins, refer to the datasheet for reconfiguration.

---

7. References

- ESP32-C3-WROOM-02 Module Datasheet

  https://documentation.espressif.com/esp32-c3-wroom-02_datasheet_en.pdf

- ESP32-C3 SoC Datasheet

  https://www.espressif.com/sites/default/files/documentation/esp32-c3_datasheet_en.pdf

- ESP32-S3 Development Solutions

https://documentation.espressif.com/en/documentList?c=ACS+Solution&c=ACK+Solution&c=ESP+Smart+Switch+Solution&c=ESP-Skainet&c=ESP+HMI+Smart+Displays&c=ESP-DSP&c=ESP-Jumpstart&c=ESP-TOUCH&c=ESP-SR&c=ESP-NOW+latest&c=ESP-NOW+v2.0&c=ESP-WHO+latest&c=ESP-WHO+v1.1.0&c=ESP-IoT-Solution+latest&c=ESP-IoT-Solution+v2.0&c=ESP-Drone&c=ESP-Brookesia&c=ESP-DL&c=ESP-BOX&c=ESP-IoT-Bridge&c=ESP32-Camera&s=ESP32-S3&eol=false

- Native USB driver installation guide

https://docs.espressif.com/projects/esp-techpedia/en/latest/esp-friends/get-started/try-firmware/try-firmware-usb-driver.html

https://docs.espressif.com/projects/esp-iot-solution/en/latest/usb/usb_overview/usb_serial_jtag.html#id1

---

8. Arduino IDE programming

1. Download the latest version of Arduino IDE.

2. Include the ESP board package link given below in the “Additional board manager URL under File-Preferences”:

“https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_dev_index.json”

2. As next step, install the esp32 library from the “Tools-Board-Board Manager” by typing esp32 in the search box and choosing the latest library version.

3. Once ESP32 library is installed, connect the development board and choose the **“ESP32 C3 Dev Module”** under “Tools – Board – esp32”.

4. Upload the code and start working with the board.
